

Answer Key — Biology Grade 11

Sub-Strand 1.1: Taxonomy I • Lesson 6: Applying Binomial Nomenclature: Naming Our Local Organisms the Same Way Everywhere

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	A	B	A	C	A	D

USE AFTER: PREDICT PHASE — Predict: organism cards and the prediction 'what name could they both use so nobody is confused?'

Q1. In Kenya, maize is called 'mahindi' in Kiswahili, 'mbembe' in Kikuyu and 'oganda' in Luo. Why is this a problem for scientists who share information with scientists in other countries?

- A. Only English names can be used for plants
- B. Local names are wrong and must never be used
- C. Different languages use different names, so people may not know they are talking about the same organism**
- D. Maize has too many parts to be named

Correct answer: C. The same organism has many local names, so a name in one language may not be understood elsewhere. A universal name is needed. Local names are not wrong; they are just not shared by everyone.

USE AFTER: OBSERVE PHASE — Observe: Jigsaw, rule (2) genus starts with a capital, species with a small letter

Q2. A learner types the scientific name of maize. Which way follows the rules for writing a scientific name (ignoring italics)?

- A. Zea mays**
- B. Zea Mays
- C. zea mays
- D. zea Mays

Correct answer: A. A scientific name has two parts. The genus starts with a capital letter and the species starts with a small letter, so it is written 'Zea mays'.

USE AFTER: OBSERVE PHASE — Observe: Jigsaw, rule (3) italics when printed, underline each part separately when handwritten

Q3. You are writing the scientific name of the goat, *Capra hircus*, by hand in your exercise book. How should you show that it is a scientific name?

- A. Draw one long line under both words together
- B. Underline *Capra* and *hircus* separately, each with its own line**
- C. Put the name in brackets and do not underline
- D. Write both words in capital letters

Correct answer: B. Typed names are in italics. When handwritten, the genus and the species are each underlined separately.

USE AFTER: OBSERVE PHASE — Observe: group table matching scientific names to the organism cards

Q4. A group is matching scientific names to organism cards. Which organism is correctly matched with the scientific name *Phaseolus vulgaris*?

- A. Bean plant**
- B. Mosquito
- C. Goat
- D. Maize

Correct answer: A. *Phaseolus vulgaris* is the scientific name of the bean plant. *Zea mays* is maize, *Capra hircus* is the goat and *Anopheles gambiae* is the mosquito.

USE AFTER: EXPLAIN PHASE — Explain: worksheet 'Fix the Names'

Q5. A learner writes the scientific name of the goat as 'Capra Hircus'. What is wrong with it?

- A. Both words should start with a small letter
- B. The genus should start with a small letter
- C. The species 'Hircus' should start with a small letter**
- D. Scientific names should be written in English

Correct answer: C. Only the genus begins with a capital letter. The species starts with a small letter, so the correct name is 'Capra hircus'. This is a common error of capitalising the species.

USE AFTER: EXPLAIN PHASE — Explain: Claim-Evidence-Reasoning response on why Zea mays is a better name

Q6. A learner writes a Claim-Evidence-Reasoning answer claiming that Zea mays is a better name for scientists than mahindi, mbembe or oganda. Which reasoning best supports the claim?

- A. The same two-part name is used worldwide, so everyone can identify the same organism**
- B. Zea mays is shorter and easier to say than the local names
- C. Local names are incorrect and should not be used
- D. Zea mays is older than all the local names

Correct answer: A. A scientific name is the same everywhere, so scientists in any country know which organism is meant. Local names differ by language, but they are not wrong.

USE AFTER: EXPLAIN PHASE — Explain: discussion and debate on 'Local names are not important when we have scientific names'

Q7. In a class debate, a learner says: 'Local names are not important when we have scientific names.' Which response is best?

- A. Agree, because local names are always wrong
- B. Disagree, because local names are never used in science
- C. Agree, because scientific names replace local names
- D. Disagree, because local names such as mbembe and oganda have cultural and practical value, while scientific names allow worldwide communication**

Correct answer: D. Scientific names do not replace local names. Local names carry cultural and practical value in communities, and scientific names let people everywhere communicate about the same organism.